

Eigenvalues and inverse problems

Unitary-orbit distance under finite block repetition

Difficulty: challenging **Importance:** interesting to specialist**Status:** Partially resolved

Rating rationale: Challenging reflects preservation of a nontrivial spectral-norm orbit distance under larger unitary mixing; specialist impact concerns finite matrix amplification and operator-algebraic nearness.

Further partial result: first-order spectral splitting — 14 September 2026

Sidney Holden (Center for Computational Biology, Flatiron Institute, Simons Foundation). The [report](#) proves first-order finite-amplification invariance under independent spectral splitting about fixed repetitions of the certified three-point pair. The squared amplification gain is nonnegative and at most $C_{r,k}t^{3/2}$ for sufficiently small nonnegative t . The constants and radius can depend on dimensions and splitting directions. This is not exact equality at nonzero perturbation and is not uniform in amplification. General normal pairs remain unresolved; status stays Partially resolved.

See the [independent informal Codex AI-agent review](#) for the exact audited scope and checks, and the [submission record](#) for verified affiliation, original materials and reproduction details. No full resolution, external human peer review, formal verification or historical novelty is claimed. No Lean verification was performed.

Further partial result — 12 September 2026

Sidney Holden (Center for Computational Biology, Flatiron Institute, Simons Foundation) proves in [Theorem 1 and its amplification corollary](#) that the spectral-norm unitary-orbit distance equals the bottleneck spectral-matching distance when at least one normal input has at most two distinct eigenvalues. The explicit matching-capacity conditions are unchanged by finite block repetition, proving the displayed equality for this subclass, with arbitrary multiplicities and an arbitrary normal spectrum for the other input.

The argument passed an [independent informal Codex AI-agent audit](#). Status remains Partially resolved: normal pairs with at least three distinct eigenvalues in each spectrum remain unresolved. [Submission and verified affiliation](#). No novelty, external human peer review or formal verification is claimed.

Problem statement

For $A, B \in \mathbb{C}^{n \times n}$, define the distance between their unitary similarity orbits in the spectral norm by

$$\delta_n(A, B) = \min_{U \in \mathbb{C}^{n \times n}, U^*U = I_n} \|A - U^*BU\|_2.$$

For a positive integer k , write $A^{(k)} = I_k \otimes A$, the block diagonal matrix with k copies of A . Is it true that, for every $n \geq 3$, every finite integer $k \geq 2$, and every pair of normal matrices $A, B \in \mathbb{C}^{n \times n}$,

$$\delta_{nk}(A^{(k)}, B^{(k)}) = \delta_n(A, B)?$$

Normality means $AA^* = A^*A$ and $BB^* = B^*B$. The minimizing unitary on the left is unrestricted in dimension nk ; it may mix different copies. Thus the question asks whether block repetition can improve the best unitary-similarity fit between normal matrices.

Why it matters

This is a matrix-nearness problem with prescribed spectra. It tests whether a structured enlargement changes the optimum in spectral norm, a metric for which matching eigenvalues alone need not give the orbit distance.

References

- L. W. Marcoux and Y. Zhang, *On Specht's Theorem in UHF C-algebras**, Journal of Functional Analysis 280 (2021), 108778, §5, Question 5.4, pp. 26–27; [institutional full text](#).
- L. W. Marcoux, P. Sarkowicz, and Y. Zhang, *Kaplansky's problem and unitary orbits in matrix amplifications*, arXiv:2508.13834v1 (2025), §3.5 and Propositions 3.15, 3.21, and 4.14.

Earlier status check — 2026-09-08

Theorem 5.3 of the source proves equality for normal matrices of order two. Theorem 5.1 establishes a strict decrease for some unrestricted matrices, so normality cannot simply be omitted. The example after Question 5.4 uses infinitely many copies and does not resolve the finite- k question. The direct follow-up 2508.13834v1 records the order-two result after Corollary 3.5. The corollary itself proves equality for self-adjoint matrices of every order. Its normal-element Propositions 3.15 and 3.21 concern zero orbit distance; its Proposition 4.14 counterexamples have no normality requirement. Neither settles preservation of all distances for finite normal matrices. The published §5 and these later passages were checked. Searches for *Marcoux Zhang Question 5.4 normal matrices*, *normal matrices ampliations unitary orbit distance*, and 2025/2026 follow-ups found no finite-dimensional normal counterexample or general proof. This is limited search evidence, not a certificate of openness.

Audit update — 2026-09-10

Rechecked [Marcoux–Sarkowicz–Zhang](#), Corollary 3.5: equality is proved for every pair of self-adjoint matrices, in every dimension and finite multiplicity. This substantive normal subclass was missing from the earlier status explanation. General normal matrices remain unresolved in the checked sources; finite-normal amplification searches found no completion. The nonnormal counterexamples and zero-distance results do not settle the remaining positive-distance question.